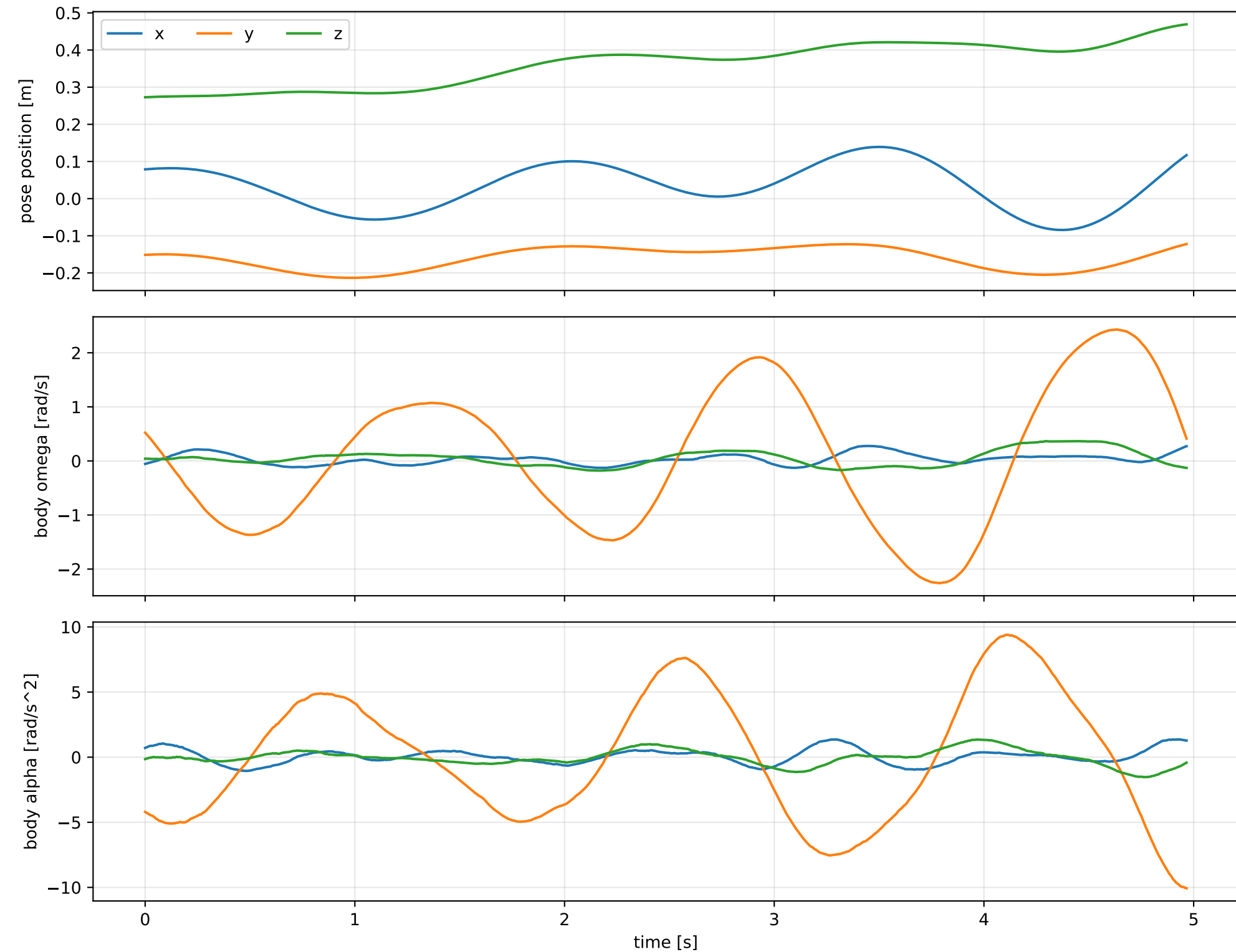
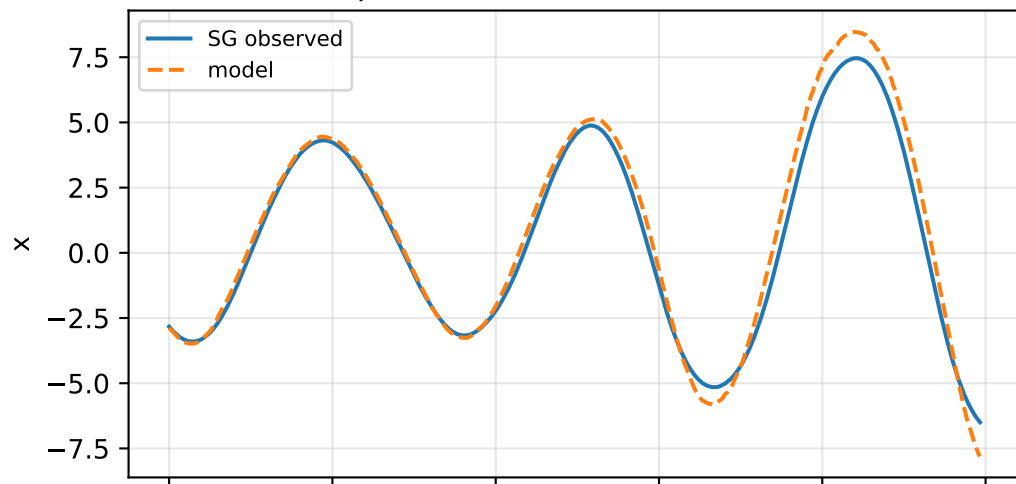


lag_fixed_one_period: SG trajectory and derived motion

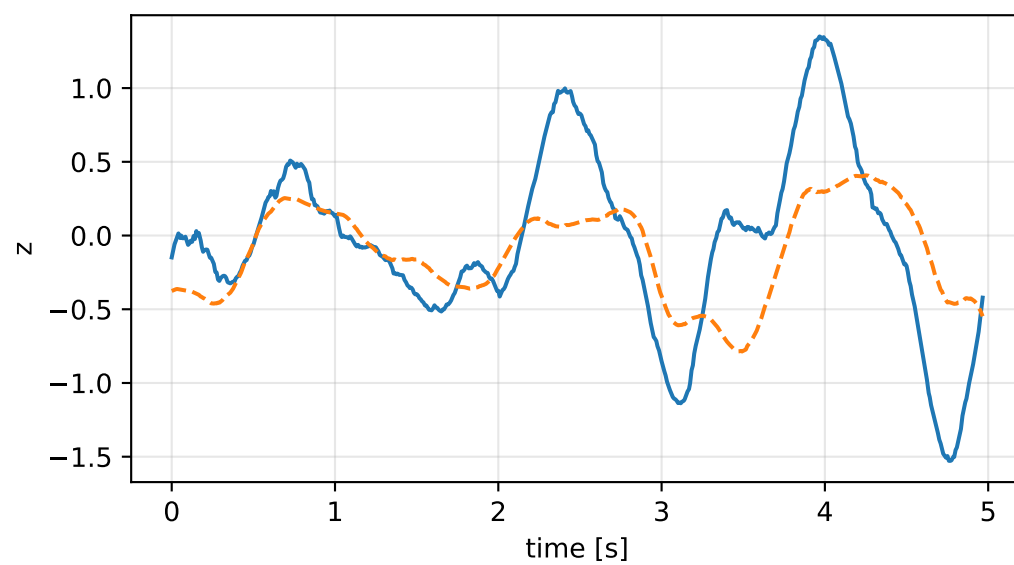
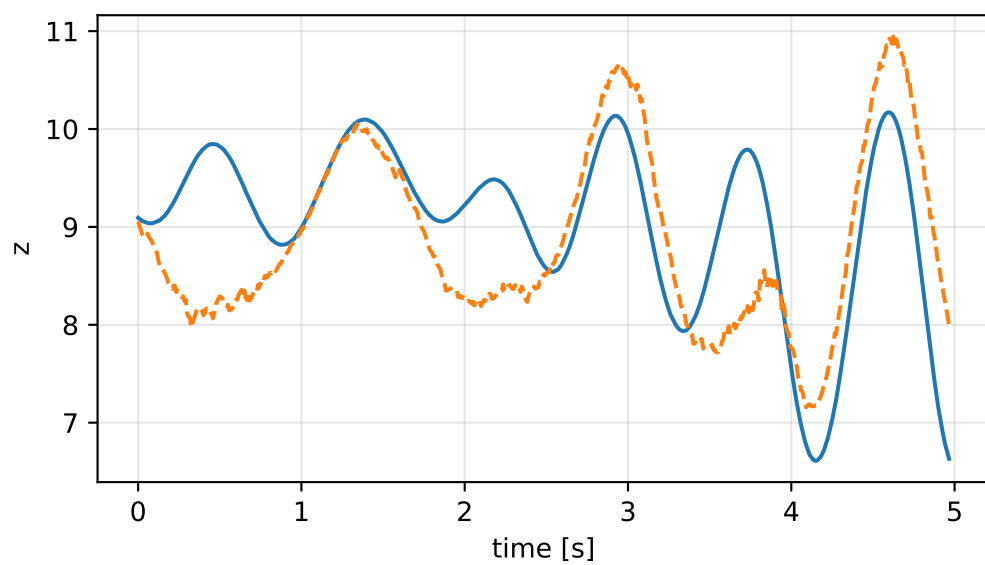
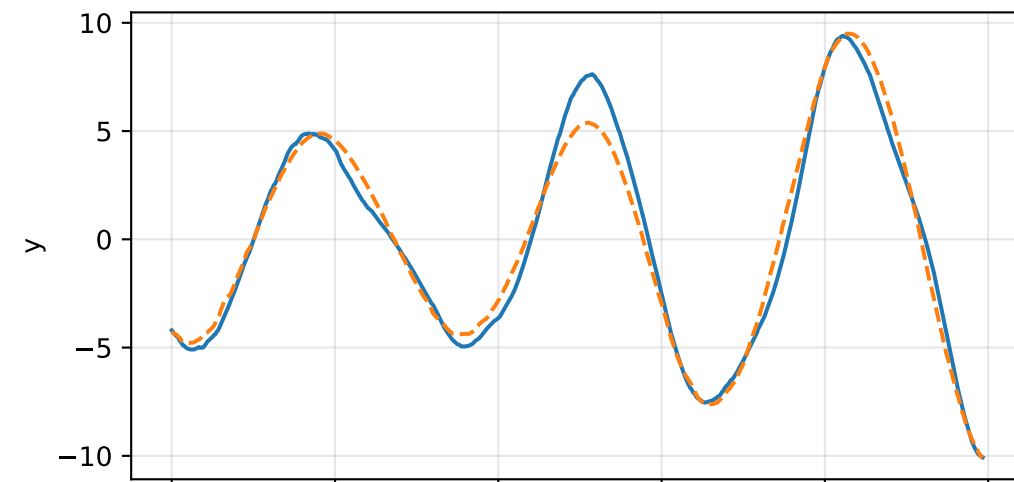
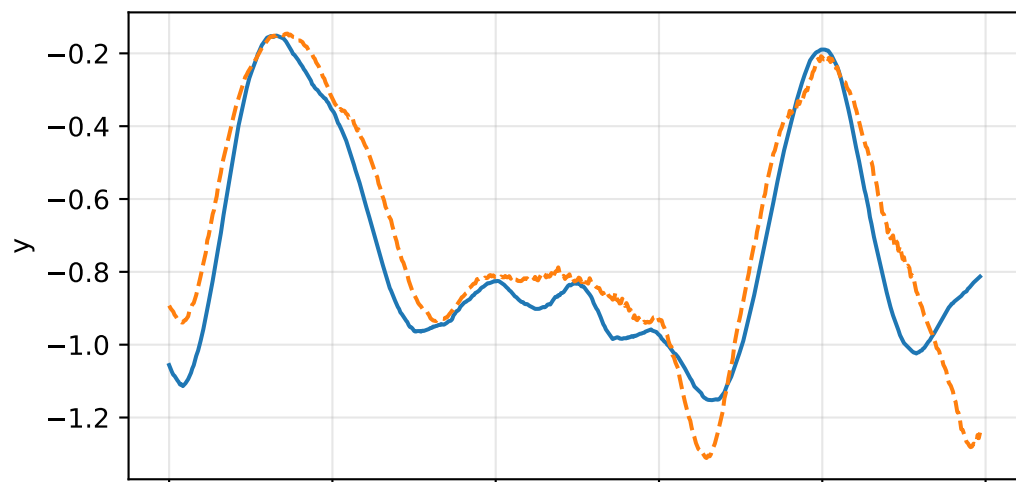
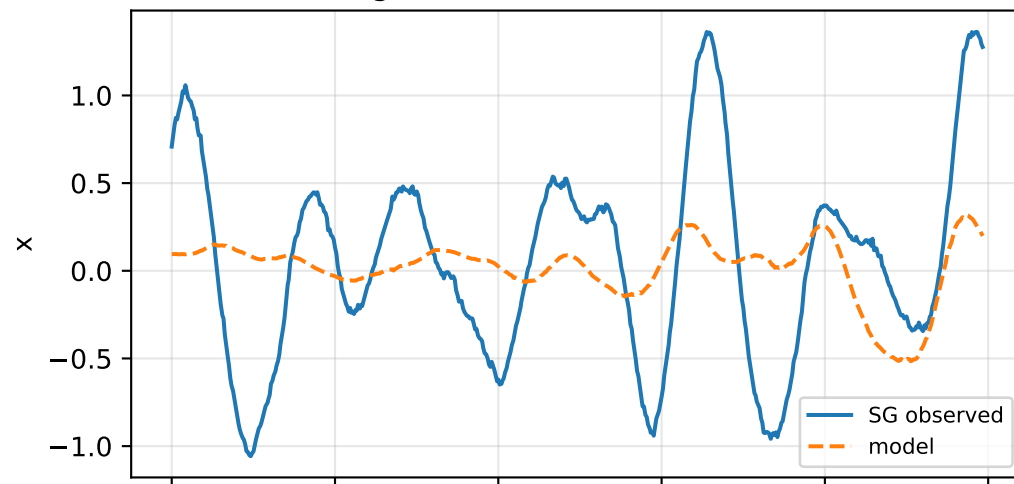


lag_fixed_one_period: acceleration residual objective

specific acceleration [m/s²]

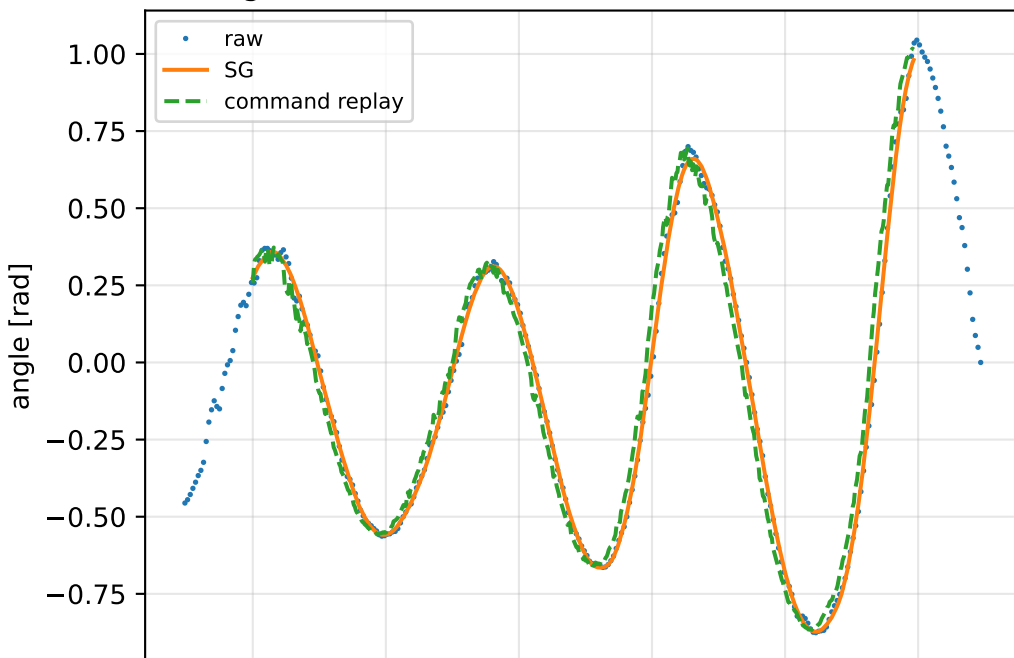


angular acceleration [rad/s²]

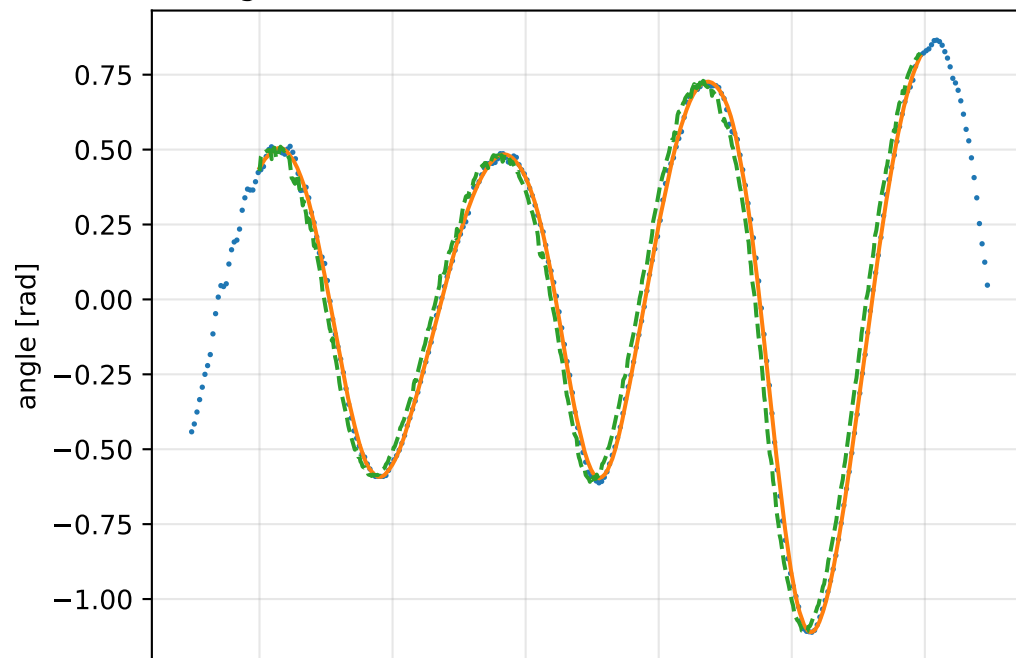


lag_fixed_one_period: gimbal measurement audit (objective=measured_sg)

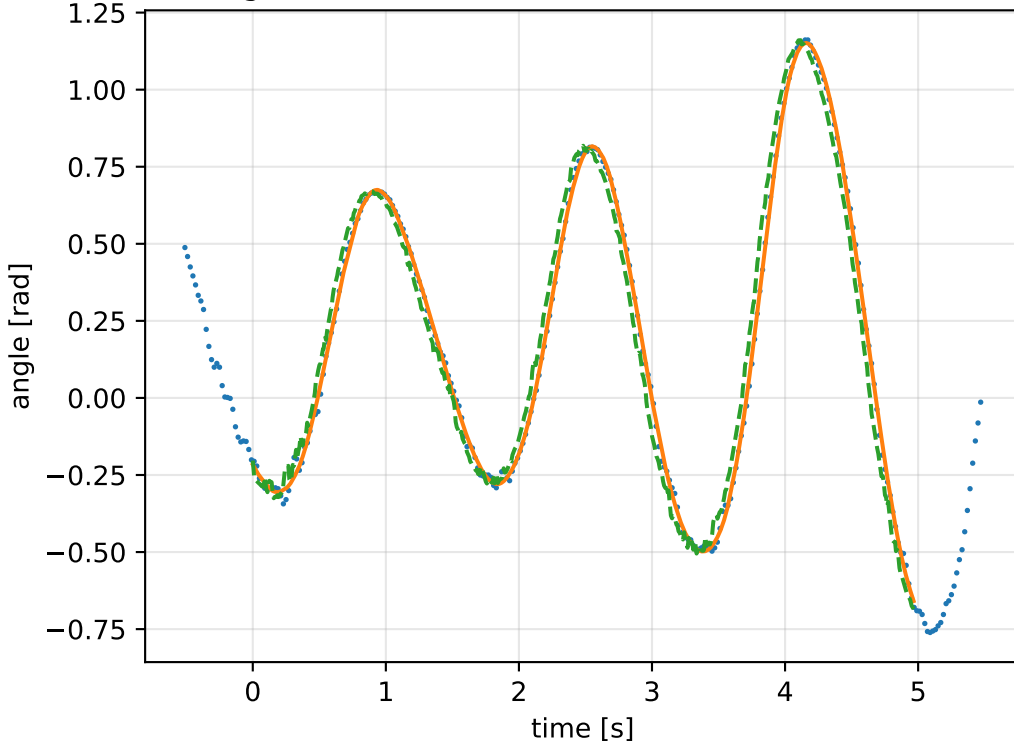
gimbal 1: RMSE=0.0759, max=0.2095 rad



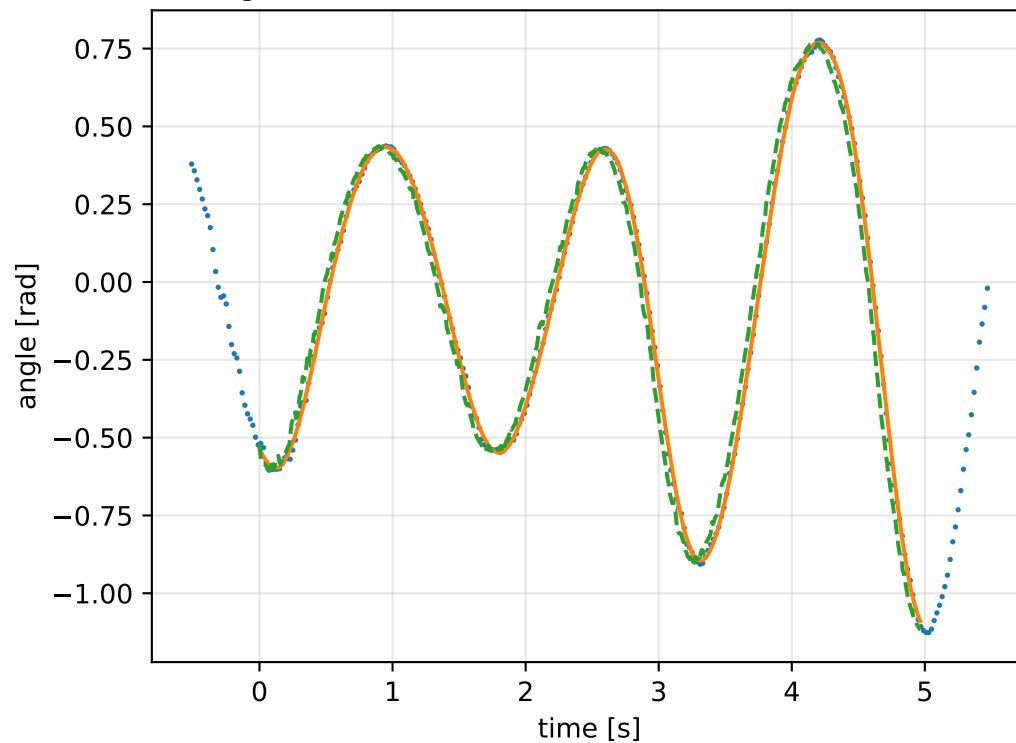
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

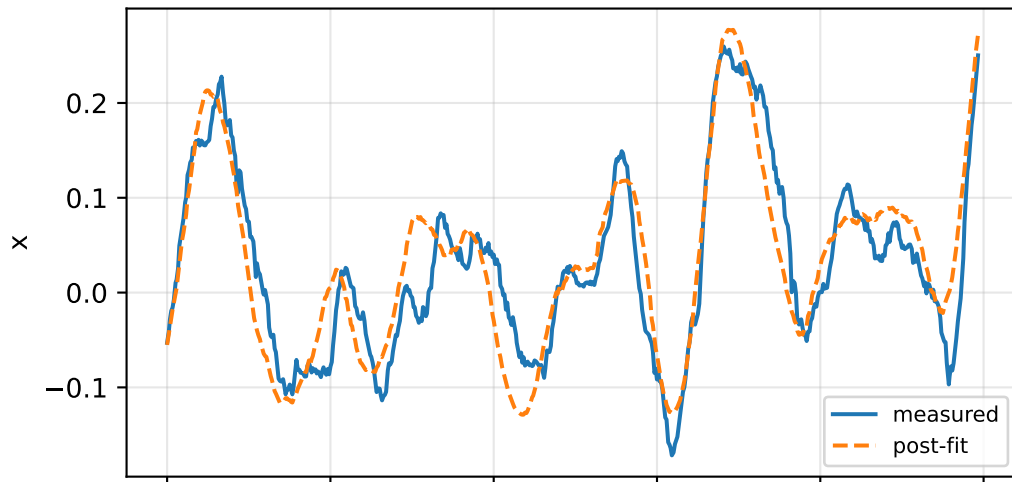


gimbal 4: RMSE=0.07131, max=0.1737 rad

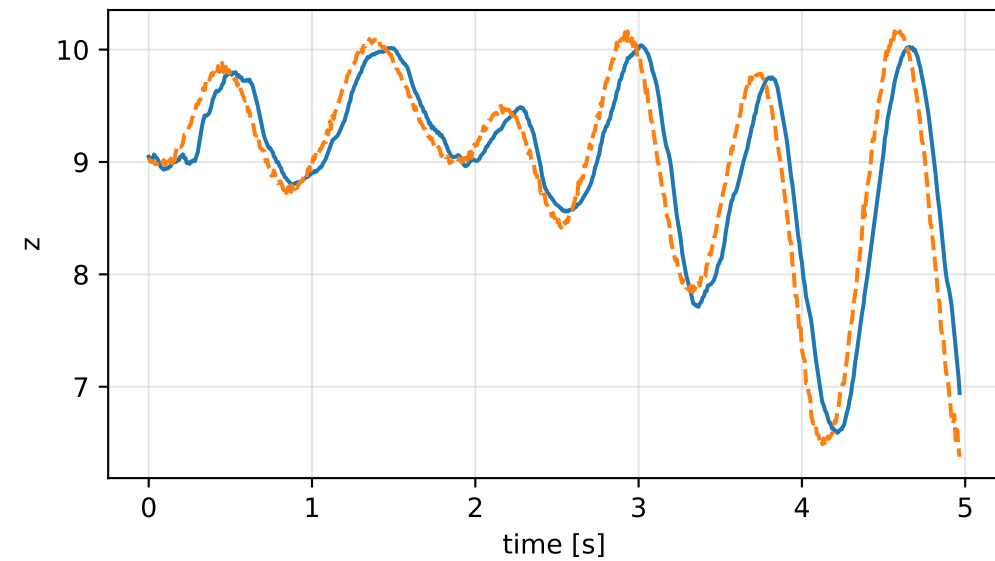
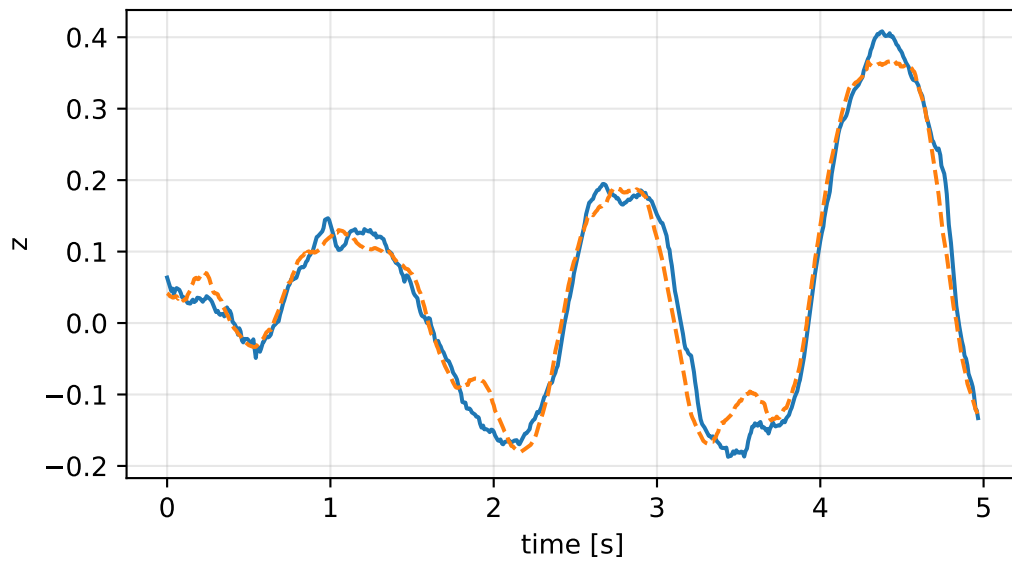
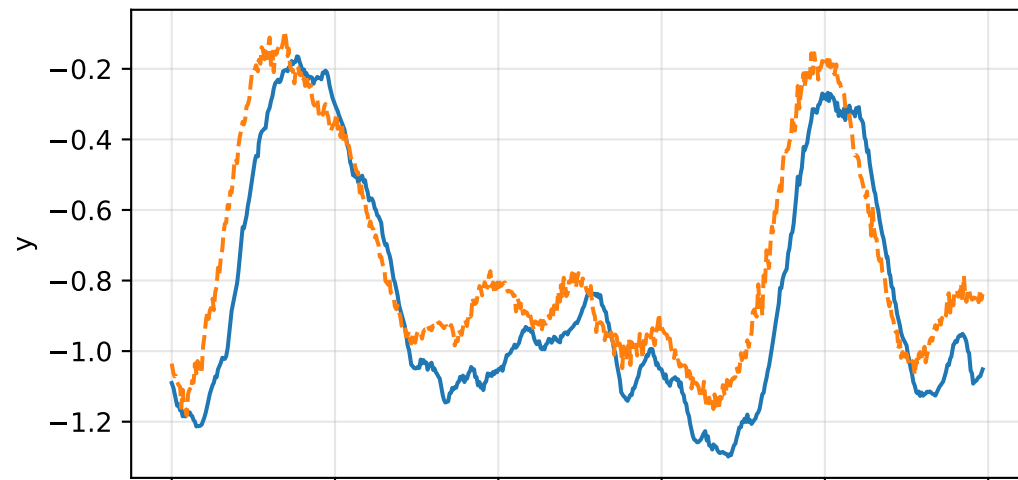
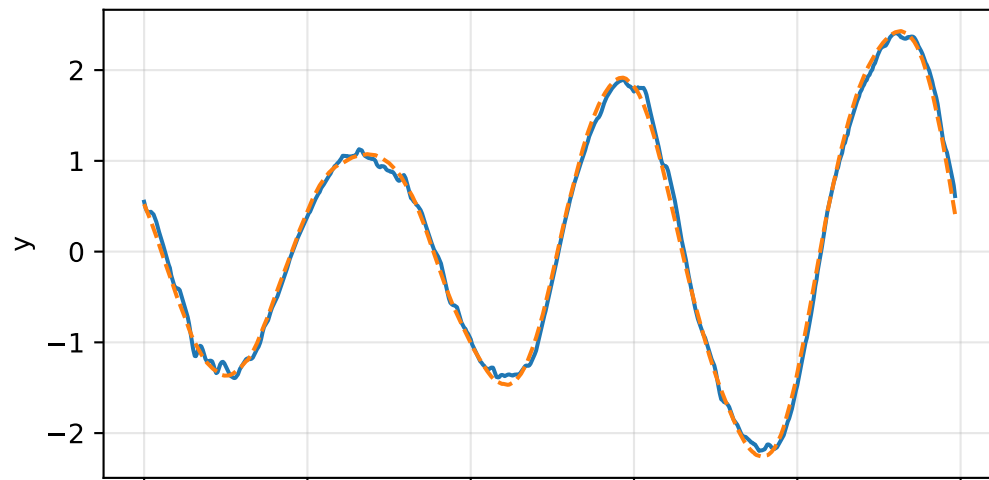
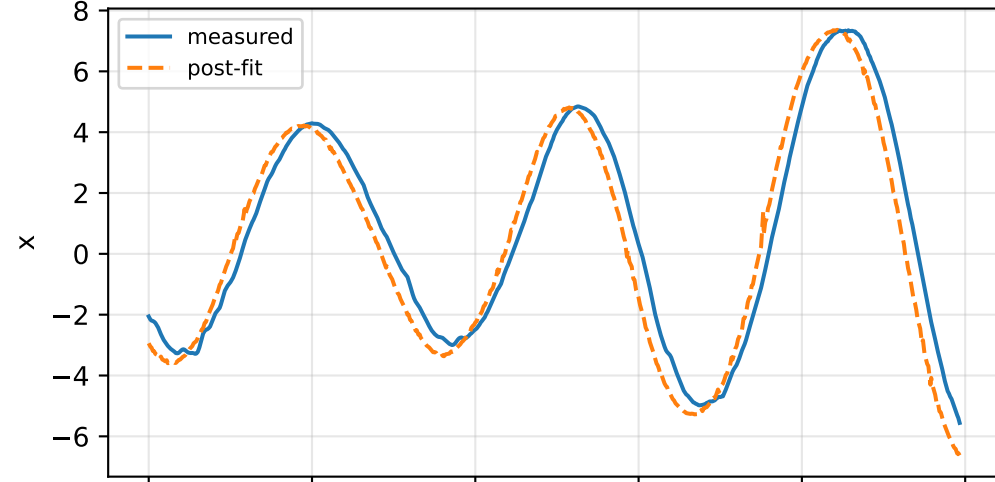


lag_fixed_one_period: IMU comparison (diagnostic only)

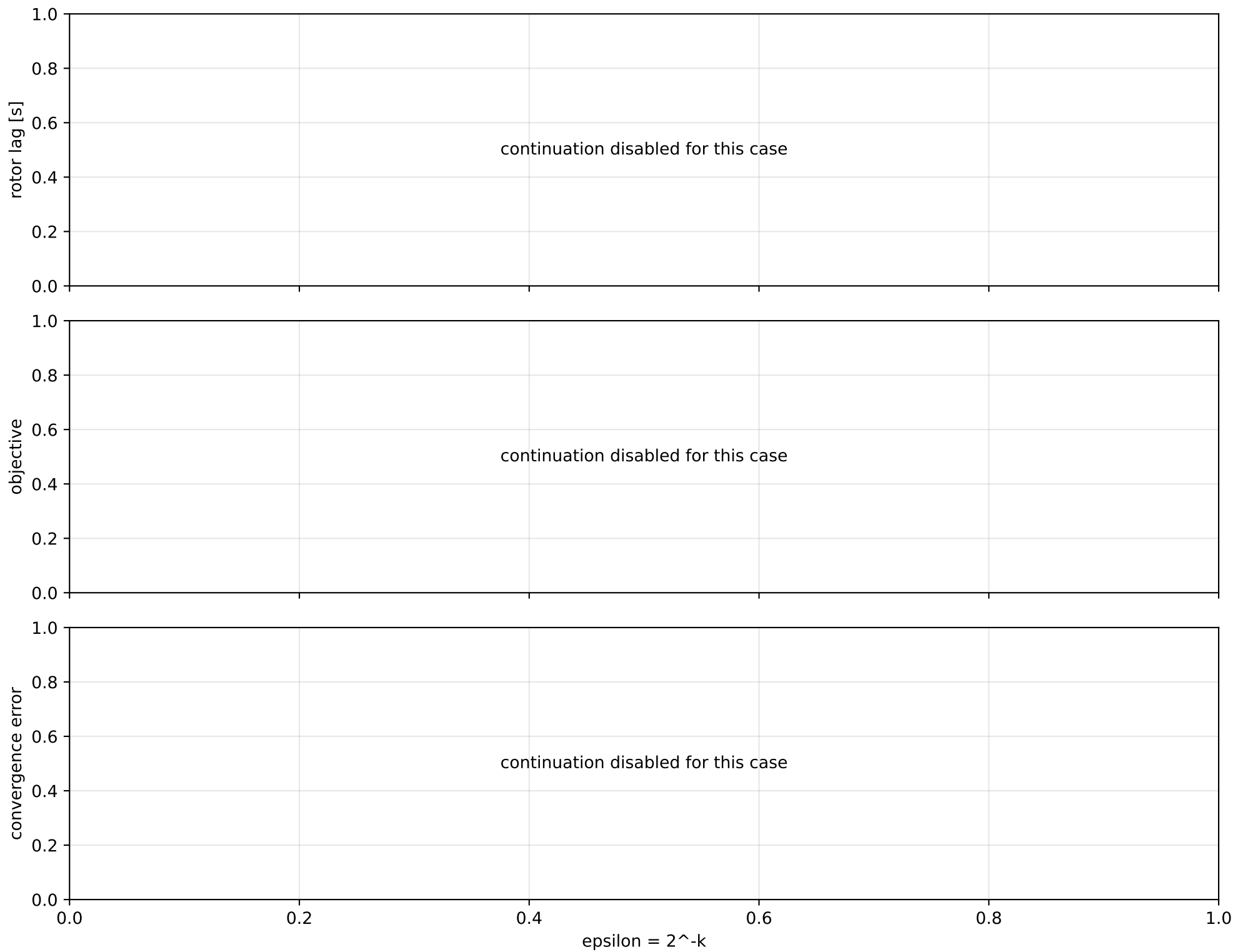
gyro [rad/s]



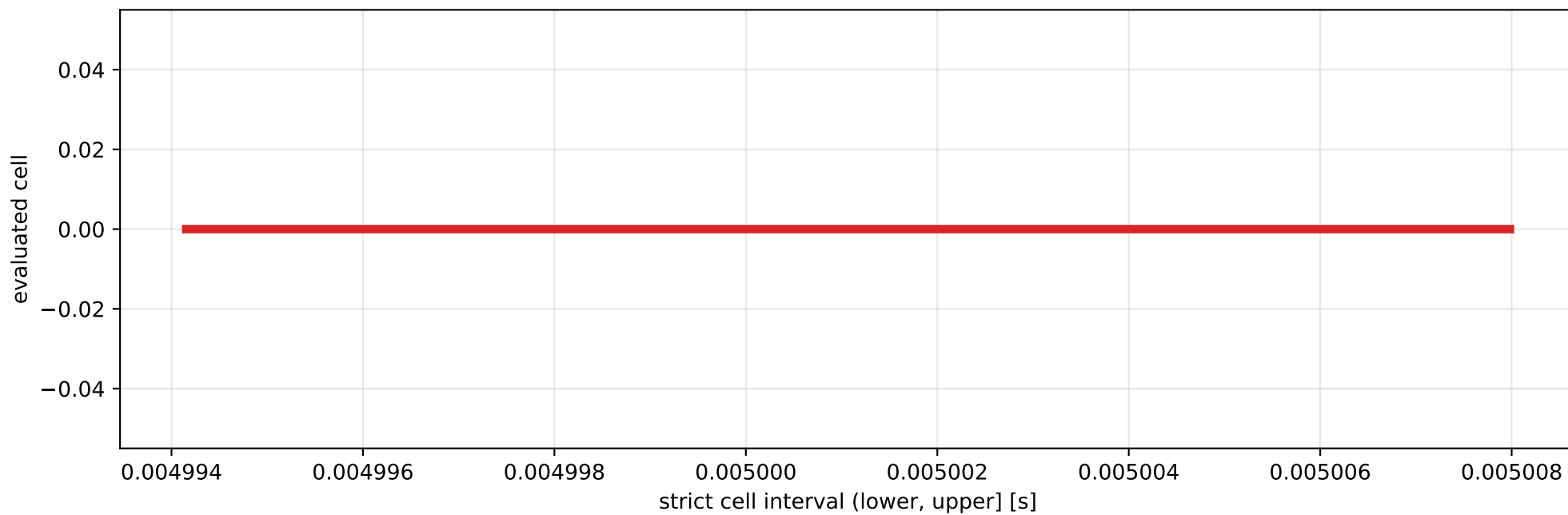
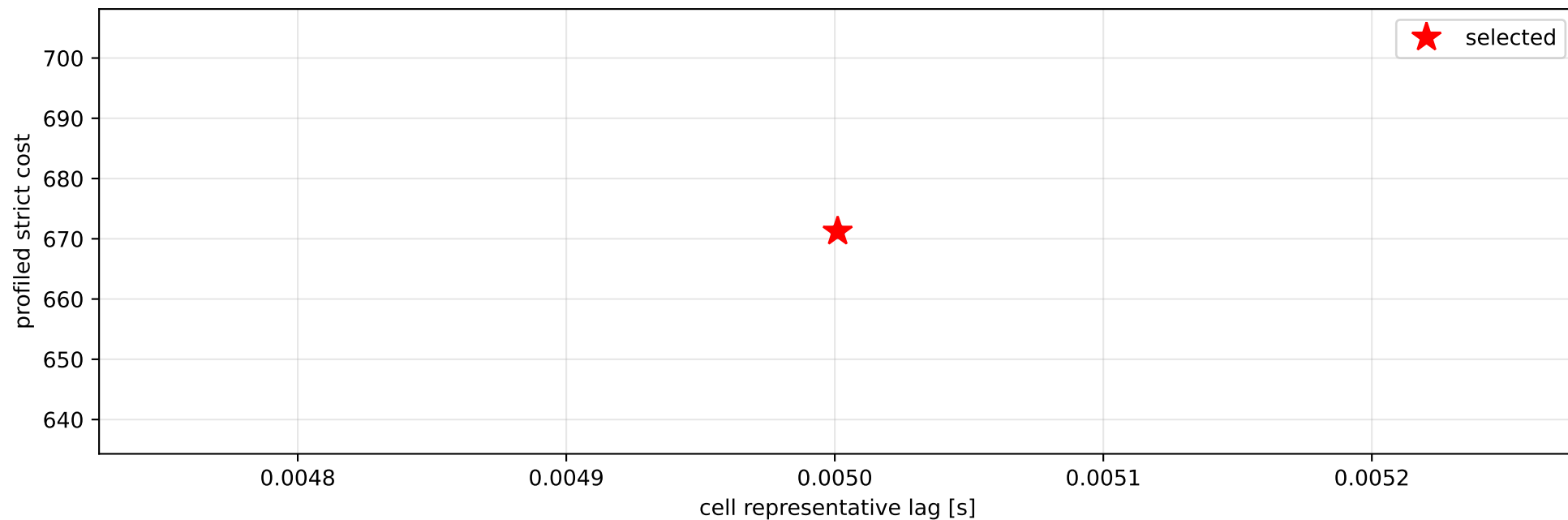
specific force [m/s²]



lag_fixed_one_period: smooth-to-strict continuation



lag_fixed_one_period: exact strict-ZOH lag cells



selected (0.00499415398, 0.00500798225] s; final smooth 0.00499773026 s; data boundary=False

lag_fixed_one_period: parameter estimate

Case: lag_fixed_one_period

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 6.00423593

inertia [kg m²]:

```
[[ 8.65798335  0.0591329  0.06518215]
 [ 0.0591329  0.17550657 -0.45689193]
 [ 0.06518215 -0.45689193  8.54951111]]
```

CoG body [m]: [0.21468599 -0.00495763 -0.04751993]

force effectiveness: [7.69787664e-04 4.49441135e+00 4.11779510e+00 1.46394539e-01]

scale-free J/m [m²]:

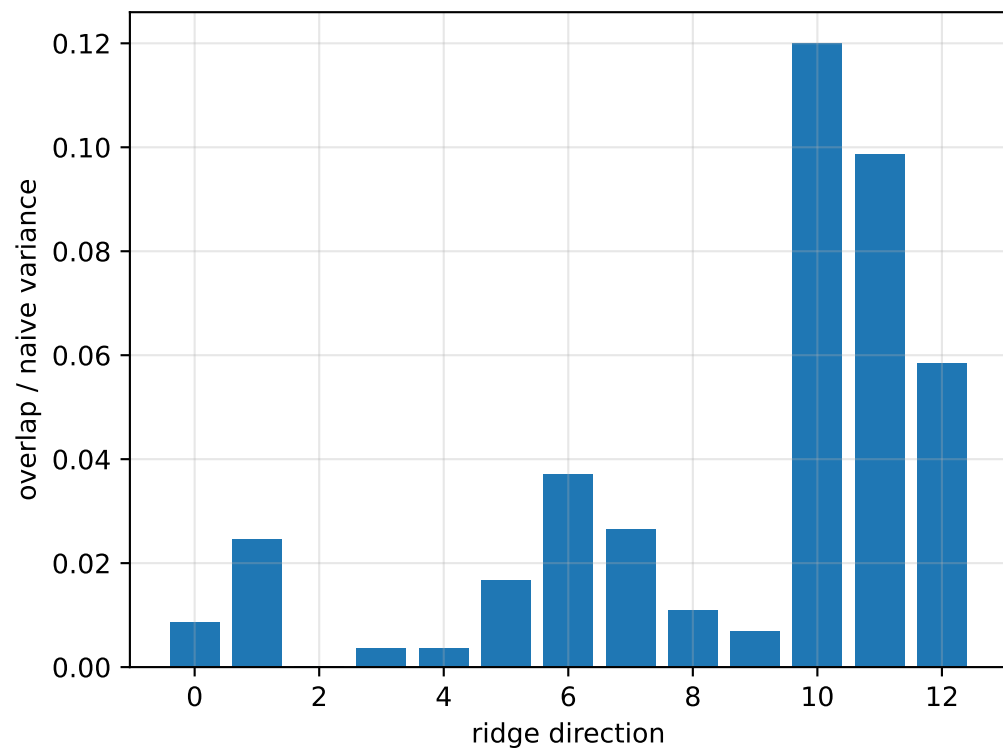
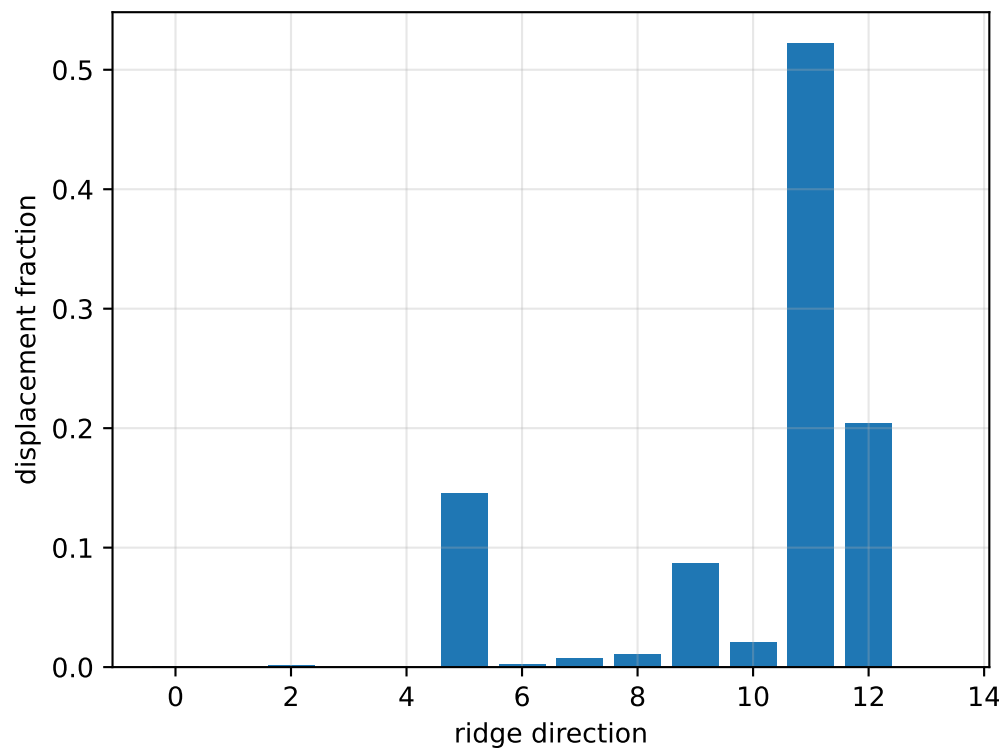
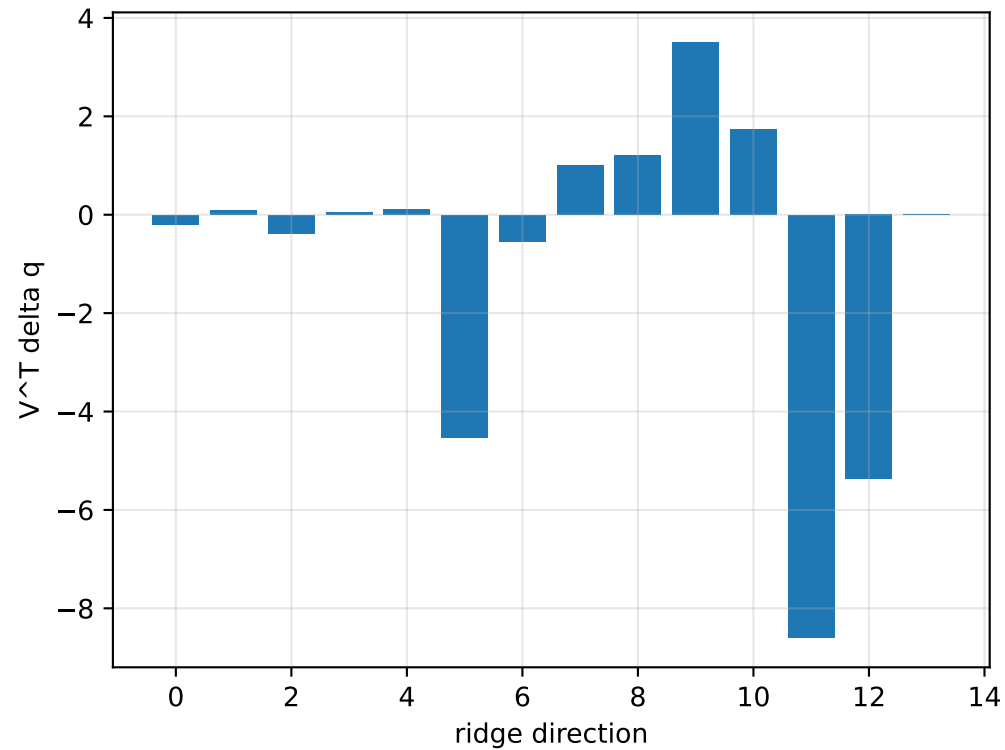
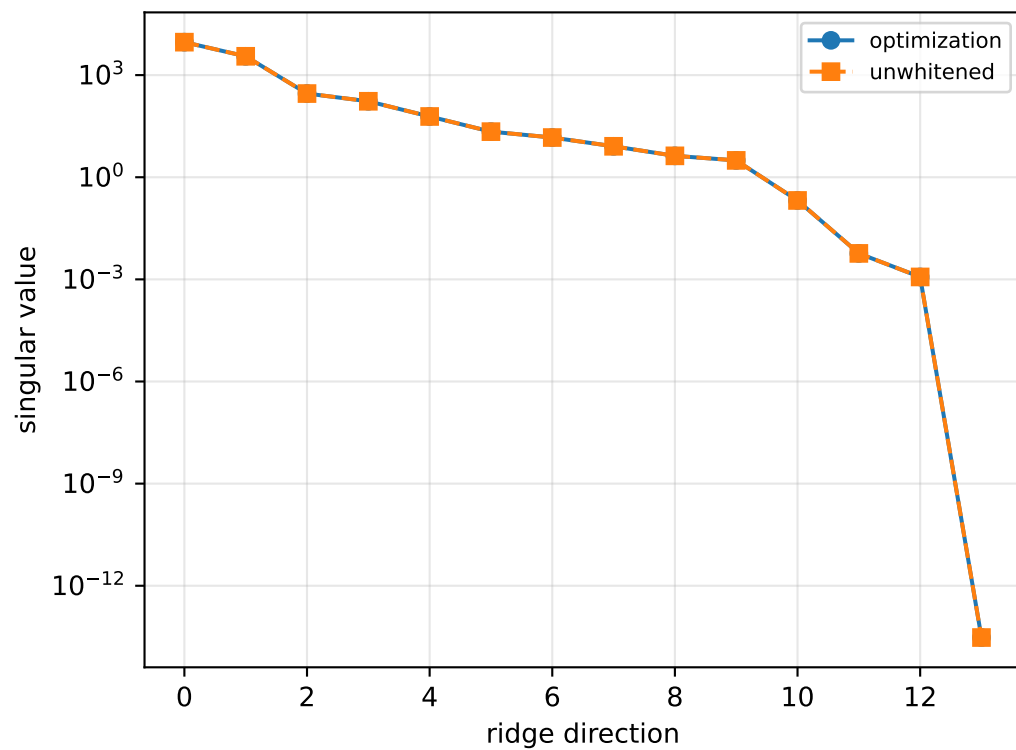
```
[[ 1.4419792  0.00984853  0.01085603]
 [ 0.00984853  0.02923046 -0.07609493]
 [ 0.01085603 -0.07609493  1.42391325]]
```

scale-free f/m: [1.28207431e-04 7.48540097e-01 6.85815006e-01 2.43818765e-02]

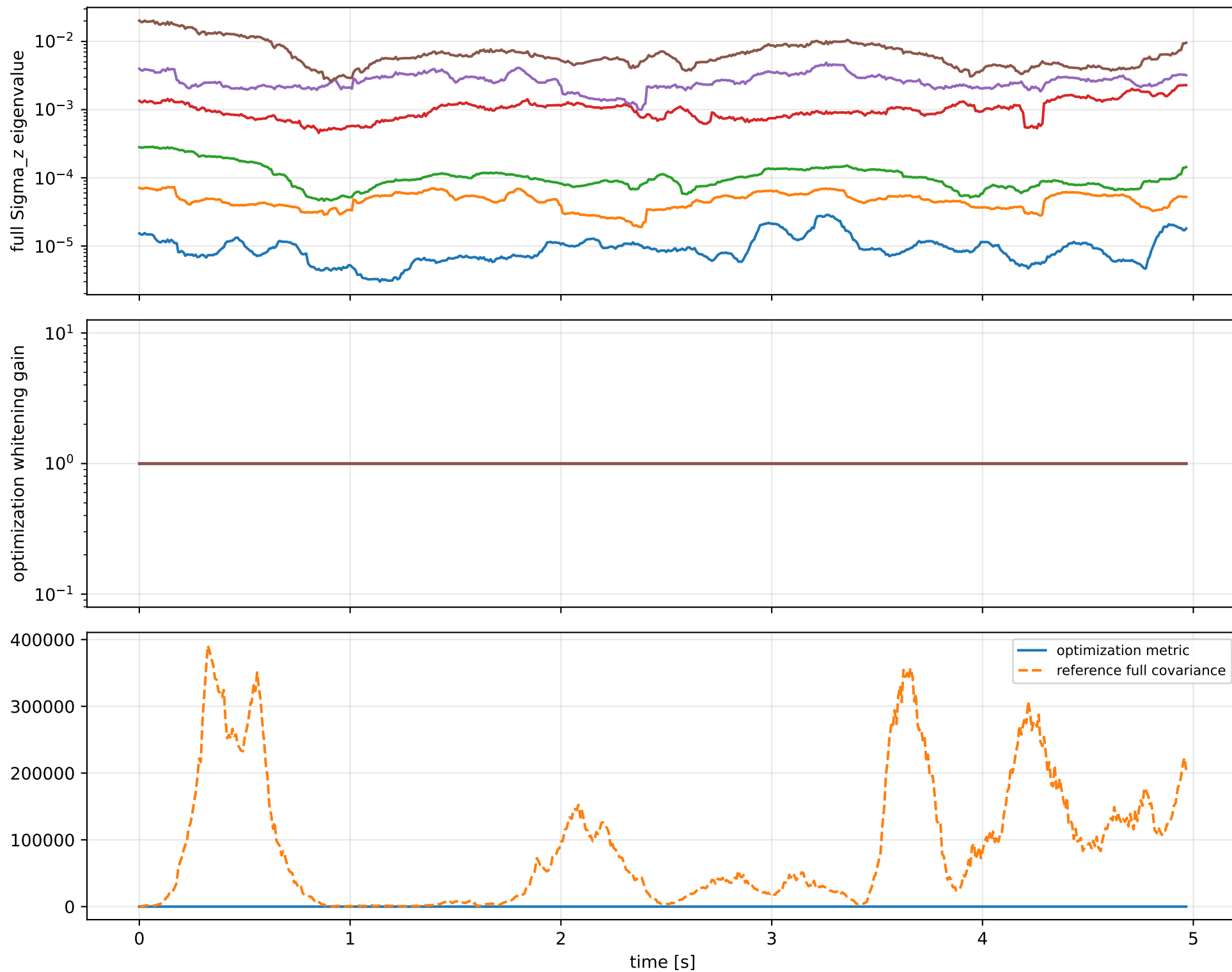
rotor lag cell: (0.00499415398, 0.00500798225) s

representative: 0.00499773026 s

lag_fixed_one_period: ridge spectrum (rank 13, nullity 1, $||jv||=1.73e-13$)



lag_fixed_one_period: optimization vs reference covariance



lag_fixed_one_period: wrench / closure (force max 1.69e-14, torque max 3.55e-15)

